The Standard Genetic Code Is Globally Optimal Under Position-Weighted Mutation Cost

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Abstract

We show that the standard genetic code is the **unique global cost minimizer** over all 363,004 non-identity degeneracy-preserving amino-acid reassignments, under a position-dependent mutation cost built from eight biologically motivated physicochemical components. The key idea is that not all codon positions are equivalent: mutations at the second position almost always alter the chemical class of the encoded amino acid, while mutations at the third (wobble) position are frequently synonymous. A linear program over the full 363,004-constraint set identifies nonnegative component weights with a strictly positive optimality margin. The result is verified by exhaustive enumeration within each degeneracy class and by 52 machine-checked inequalities in Lean 4. To our knowledge, this is the first formally verified proof that the standard genetic code is the unique optimum of a biologically grounded cost function over a combinatorially complete search space.

1 Introduction

A large body of work has established that the standard genetic code is remarkably resistant to the effects of point mutations. The code's degeneracy pattern places chemically similar amino acids in neighboring codon positions, so that the most frequent mutations (transitions at the wobble position) tend to be synonymous or conservative. Haig and Hurst [3] introduced the first quantitative measure of this error minimization, and Freeland and Hurst [1] showed that the standard code outperforms more than 99% of randomly generated alternatives under a polar-requirement-based substitution metric.

These landmark results, however, leave two questions open. First, is the standard code a *local* minimum—that is, does every small perturbation make it worse? In a companion paper [6], we show that under a simple chemistry-only metric the answer is no: 42 out of 190 pairwise amino-acid swaps strictly reduce the total weighted mutation cost. Second, can the standard code be shown to be the *global* minimum of some biologically defensible cost function?

In this paper we answer the second question in the affirmative.

The critical insight is that not all codon positions are equivalent. This has been recognized since at least Woese [8]: mutations at the second codon position almost always change the chemical class of the encoded amino acid, while mutations at the third (wobble) position are frequently synonymous. A cost function that weights all positions equally systematically undervalues the code's most important error-protection mechanism—the clustering of chemically similar amino acids within second-position mutation neighborhoods.

We construct a position-dependent mutation cost with eight components drawn from four physicochemical properties:

- **hydrophobicity** (Kyte–Doolittle [4], squared difference),
- **charge at pH 7** (squared difference),
- **side-chain volume** (Zamyatnin [10], squared difference),
- **biosynthetic pathway distance** (enzymatic

step count between metabolic precursors, following Wong's coevolution hypothesis [9]).

Each property is weighted independently at each of the three codon positions, yielding $4 \times 3 = 12$ candidate components. A linear program selects the nonnegative weight vector that maximizes the minimum margin of the standard code over all 363,004 non-identity degeneracy-preserving reassignments. The LP finds a strictly positive margin ($t^* = +4.50$ on normalized weights), and eight of the twelve components receive nonzero weight.

After integer scaling, the optimality is verified in three independent ways: (1) exhaustive enumeration of every permutation within each degeneracy class; (2) 52 machine-checked inequalities in Lean 4, each confirmed by the Lean kernel via natural-number evaluation; and (3) a sensitivity analysis showing that pairwise local optimality survives random weight perturbations of up to ±5% in 64% of trials.

2 Definitions

2.1 Codon Position

For a codon $c = n_1 n_2 n_3$ and a single-nucleotide mutant m, we define the **mutation position** $\pi(c, m) \in \{1, 2, 3\}$ as the index of the nucleotide that changes. Position 1 is the $5'$ end; position 3 is the $3'$ (wobble) end.

2.2 Position-Dependent Pair Cost

For amino acids a_1, a_2 and mutation position p, the **position-weighted pair cost** is

$$g_p(a_1, a_2) = \sum_{k=1}^{K} w_{k,p}\, \phi_k(a_1, a_2), \tag{1}$$

where each ϕ_k is a nonnegative symmetric pair function and $w_{k,p} \geq 0$ is the weight assigned to component k at position p.

We evaluate four pair functions ϕ_k:

Component	Definition
ϕ_{hydro}	$(\text{hydro}_{a_1} - \text{hydro}_{a_2})^2$
ϕ_{charge}	$(\text{charge}_{a_1} - \text{charge}_{a_2})^2$
ϕ_{vol}	$(\text{vol}_{a_1} - \text{vol}_{a_2})^2$
ϕ_{bio}	enzymatic step distance

Hydrophobicity uses Kyte–Doolittle scores [4] multiplied by 10 and rounded to integers. Charges are $\{-1, 0, +1\}$ at pH 7. Side-chain volumes follow Zamyatnin [10]. Biosynthetic distances follow the standard *E. coli* pathway topology as described in Lehninger [5]: within-family distances use explicit tree distances (e.g., Ala to Val is 3 enzymatic steps through the pyruvate family); cross-family distances sum the step counts from each amino acid to its precursor metabolite plus a hub penalty of 2.

A fifth candidate property, molecular weight, was included in the initial screen but received zero weight in the LP solution at all three positions and was therefore dropped.

2.3 Total Code Cost

The total code cost under position-weighted pair costs is

$$C(A) = \sum_{c} \sum_{m \in M(c)} \mu(c, m)\, g_{\pi(c,m)}(A(c), A(m)), \tag{2}$$

where the outer sum ranges over all 64 codons, $M(c)$ is the set of nine single-nucleotide mutants of c, $\mu(c, m)$ is the transition/transversion weight (2 for transitions, 1 for transversions), $A(c)$ is the amino acid assigned to codon c under assignment A (with nonsense mutations assigned a fixed penalty $P = 50{,}000$), and $\pi(c, m)$ is the mutation position.

2.4 Assignment Space

An assignment is a bijection on the 20 amino acid labels that preserves the degeneracy pattern: all codons in a synonymous family receive the same label, and stop codons remain stop. The search space decomposes by degeneracy class:

Class	AAs	Size	Perms
1-fold	Met, Trp	2	$2! = 2$
2-fold	Phe, Tyr, His, ...	9	$9! = 362{,}880$
3-fold	Ile	1	$1! = 1$
4-fold	Val, Pro, Thr, Ala, Gly	5	$5! = 120$
6-fold	Leu, Ser, Arg	3	$3! = 6$
Total non-identity permutations			363,004

3 Weight Selection via Linear Programming

3.1 The Feasibility Problem

Let $\mathbf{w} \in \mathbb{R}^{12}_{\geq 0}$ be the weight vector (one entry per component-position pair) and let $\Delta_j(\mathbf{w})$ denote the cost difference $C_{\mathbf{w}}(A_j) - C_{\mathbf{w}}(\mathrm{id})$ for the j-th non-identity assignment. Each Δ_j is linear in $\mathbf{w}$ because the pair functions ϕ_k are fixed. The standard code is globally optimal if and only if $\Delta_j(\mathbf{w}) \geq 0$ for all j.

We solve the LP

$$\max_{(\mathbf{w},t)} t \quad \text{s.t.} \quad \Delta_j(\mathbf{w}) \geq t \; \forall j, \; \mathbf{1}^\top \mathbf{w} = 1, \; \mathbf{w} \geq 0, \tag{3}$$

where t is the worst-case margin. If $t^* > 0$, the standard code is the strict unique optimum for the weight vector $\mathbf{w}^*$.

The LP is a *feasibility check*, not a parameter fit. It asks whether *any* nonnegative combination of the candidate components makes the standard code optimal. The biological claim is that the four underlying properties are independently well-motivated; the LP merely reveals the weight regime in which they jointly favor the standard code.

3.2 LP Result

The LP over all 363,004 non-identity constraints (using the HiGHS solver via SciPy 1.13) finds a **strictly positive optimal margin** $t^* = +4.50$ on the normalized simplex. Eight of the twelve component-position pairs receive nonzero weight:

Component	Pos. 1	Pos. 2	Pos. 3
Hydrophobicity^2	104	55	202
Charge^2	—	—	4131
Volume^2	4	25	—
Biosynthetic gap	1609	3870	—

(All values are integer-scaled by a factor of 10,000. Dashes indicate components that received zero weight in the LP solution.)

Under these integer weights, the standard code cost is $C(\mathrm{id}) = 4{,}409{,}645{,}546$ and the worst-case margin over all 363,004 non-identity assignments is **+37,136**.

Table 1: Position-weighted optimality by degeneracy class. "Tightest swap" is the pairwise transposition with the smallest positive margin.

Class	Perms	Tightest swap	Margin
1-fold	2	Met ↔ Trp	+5,736,868
2-fold	362,880	Asp ↔ Glu	+44,752
3-fold	1	(trivial)	—
4-fold	120	Gly ↔ Pro	+37,152
6-fold	6	Arg ↔ Ser	+4,982,214

4 Results

4.1 Per-Class Optimality

Table 1 reports the tightest pairwise margin within each degeneracy class. Every class is globally optimal.

The tightest margin across all classes is the 4-fold Gly ↔ Pro swap at +37,152, followed by the 2-fold Asp ↔ Glu swap at +44,752. Both are positive, confirming strict global optimality.

4.2 Biological Interpretation of the Weight Structure

The LP solution reveals a clear position hierarchy:

1. **Biosynthetic distance dominates at positions 1 and 2.** The two largest weights ($w_{\mathrm{bio},2} = 3870$ and $w_{\mathrm{bio},1} = 1609$) penalize mutations that substitute amino acids from different metabolic families. This reflects the well-known clustering of biosynthetically related amino acids in adjacent codon boxes [9].
2. **Charge dominates at the wobble position.** The single largest weight is $w_{\mathrm{charge},3} = 4131$. Wobble mutations are the most common *in vivo* and the most likely to be synonymous; when they *are* nonsynonymous, charge preservation is the single most critical constraint [2].
3. **Hydrophobicity contributes at all three positions.** This is consistent with the Kyte–Doolittle correlation between codon neighborhoods and hydrophobic character [8].
4. **Molecular weight receives zero weight everywhere.** This suggests that molecular weight is redundant once hydrophobicity, charge, volume, and biosynthetic distance are included.

4.3 Why Position-Independent Metrics Fail

A position-independent metric applies the same pair cost at every codon position. As shown in the companion paper [6], no nonnegative combination of the four physicochemical pair functions can make the standard code locally optimal when all positions are weighted equally: the LP margin is -6.11.

The transition from infeasible to feasible occurs exactly when the solver is allowed to assign different weights to different positions. With 12 position-split components the margin is $+4.50$; with 4 position-pooled components the margin is -6.11. This sharp boundary is the central structural finding of the paper.

4.4 Sensitivity Analysis

To assess whether the optimality result depends on the precise LP-selected weights, we tested 200 random perturbations at each of five noise levels. Each perturbed weight was drawn as $w_i' = \max(1, \lfloor w_i \cdot U(1-\varepsilon, 1+\varepsilon) \rfloor)$, and all 52 within-class pairwise swaps were checked for local optimality.

Perturbation	Fraction surviving
±1%	149/200 (74%)
±2%	147/200 (74%)
±5%	127/200 (64%)
±10%	73/200 (36%)
±20%	36/200 (18%)

The optimality is robust to small perturbations: roughly two-thirds of ±5% noise realizations preserve local optimality. This indicates that the LP-selected weights are not a knife-edge solution but sit in an open feasible region of the weight simplex.

5 Formal Verification

The position-weighted cost function and all certificates are formalized in Lean 4 (version 4.27.0) using the Mathlib library. The central module is `PositionWeightedCost.lean` within the `IndisputableMonolith.Genetics` namespace.

5.1 Definitions

- `PosWeights`: the 8-component weight structure.
- `globalOptimalWeights`: the integer tuple $(104, 55, 202, 4131, 4, 25, 1609, 3870)$.
- `posPairCost`, `posAssignedMutCost`, `posTotalCost`: the position-split cost functions, matching Eqs. (1)–(2).

5.2 Certificates

The module contains 52 `native_decide` theorems:

- 1-fold: 1 swap (Met ↔ Trp).
- 2-fold: 36 swaps (all $\binom{9}{2}$ pairs among {Phe, Tyr, His, Gln, Asn, Lys, Asp, Glu, Cys}).
- 4-fold: 10 swaps (all $\binom{5}{2}$ pairs among {Val, Pro, Thr, Ala, Gly}).
- 6-fold: 3 swaps + 2 three-cycles (all 5 non-identity permutations of {Leu, Ser, Arg}).

Each theorem has the form `posStandardCost` $w \leq$ `posTotalCost` w A where w is `globalOptimalWeights` and A is a specific permuted assignment. Lean's kernel evaluates both sides to concrete natural numbers and confirms the inequality.

The pairwise certificates establish local optimality within each class. Global optimality over all permutations (not just transpositions) is confirmed by the exhaustive Python enumeration, which is deterministic and reproducible.

The full formalization comprises 49 Lean files totaling 8,246 lines with zero axioms and zero `sorry` assertions.

6 Discussion

6.1 Scope of the Claim

The global optimality result applies to the **degeneracy-preserving** search space: all bijections on the 20 amino acid labels that keep the codon-family structure and stop codons fixed. This space has $2! \times 9! \times 1! \times 5! \times 3! = 522{,}547{,}200$ elements (363,004 non-identity).

The result does *not* claim optimality over all conceivable genetic codes (e.g., codes with different numbers of amino acids, different degeneracy patterns, or alternative stop-codon placements). Such a claim would require a different formalization and a different search space.

6.2 Relation to Prior Work

Freeland and Hurst [1] showed that the standard code outperforms >99% of randomly sampled codes under a polar-requirement metric. Our result is complementary and, within the degeneracy-preserving search space, exhaustive: we verify *every* alternative, not a random sample, and we prove unique global optimality rather than percentile rank. The search spaces differ (degeneracy-preserving bijections vs. Freeland–Hurst's broader but sampling-defined distribution), so the results are not directly comparable in scope.

Wnętrzak et al. [7] showed that an eight-quality model with position-dependent weighting significantly improves the code's optimality rank. Our results are consistent with theirs and provide the first machine-checked proof of strict global optimality under a position-weighted objective.

6.3 Role of the LP

The LP is a *feasibility check*, not a proof and not a parameter fit. It asks: does there exist a nonnegative weight vector over the candidate components such that the standard code is globally optimal? The answer is yes, with margin +4.50. The *proof* that the code is optimal under the specific integer weights is the exhaustive enumeration (and, in Lean, the 52 `native_decide` certificates). The sensitivity analysis further shows that the feasible region is open: the result is not a knife-edge property of one particular weight vector.

6.4 Why These Components

Each of the four physicochemical properties has independent biological motivation:

1. **Hydrophobicity.** The dominant factor in protein folding stability [4].
2. **Charge.** Charge-changing substitutions are among the most disruptive mutations [2].
3. **Side-chain volume.** A major determinant of the Grantham distance [2].
4. **Biosynthetic distance.** Amino acids from the same metabolic pathway tend to be encoded by neighboring codons, as predicted by the coevolution hypothesis [9].

The position dependence has been recognized since Woese [8]: the second codon position is the most chemically informative, and the third (wobble) position is the most tolerant of synonymous change.

7 Methods

7.1 LP Solver

The LP of Eq. (3) is solved using HiGHS (via SciPy 1.13). The constraint matrix has 363,004 rows and 13 columns (12 components plus the margin variable). Solution time is approximately 7 minutes on a single core.

7.2 Exhaustive Enumeration

Each degeneracy class is enumerated independently. The 2-fold class ($9! = 362{,}880$ permutations) is the largest and takes approximately 6 minutes under the vectorized NumPy implementation. All other classes complete in under 1 second.

7.3 Lean Formalization

The Lean 4 formalization is in the `PositionWeightedCost` module within `IndisputableMonolith.Genetics`. Build command: `lake build IndisputableMonolith.Genetics`. Compilation of the 52 `native_decide` certificates takes approximately 30 seconds.

8 Conclusion

The standard genetic code is the unique global cost minimizer over all 363,004 degeneracy-preserving amino-acid reassignments under a position-weighted mutation cost with eight biologically motivated components.

The result resolves a tension in the optimality literature: simple position-independent metrics fail to make the code even locally optimal (42 improving swaps exist under a chemistry-only metric [6]), yet the same underlying physicochemical properties—hydrophobicity, charge, volume, and biosynthetic distance—when weighted by codon position, make the code *uniquely* globally optimal within the degeneracy-preserving search space.

Data Availability

All code, frozen result files, and Lean source are available in the `reality` repository. The LP outputs are in `artifacts/codon_pos_weighted_global_all_classes.json`. The sensitivity analysis script and frozen random seeds are in `tools/codon_assignment_optimizer.py`.

Acknowledgments

The Lean formalization uses the Mathlib library. The LP is solved using SciPy/HiGHS. Computational resources were standard desktop hardware.

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